

DTSA 5503

Dynamic Programming, Greedy Algorithms

FORMAT	CATEGORY	CREDITS
Online (DTSA)	Core	1
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 1 – CS Pathway	AI Engineer (27.8 / 100)	21.5 / 100
SPECIALIZATION Data Science Foundations: Data Structures and Algorithms Pathway		

22

CURRENT AVG SCORE
across 6 job roles

65

POTENTIAL SCORE
with recommended additions

+63

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

This course covers basic algorithm design techniques such as divide and conquer, dynamic programming, and greedy algorithms. It concludes with a brief introduction to intractability (NP-completeness) and using linear/integer programming solvers for solving optimization problems.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
AI Engineer	27.8	66.1	+38.3	HIGH
Applied Scientist	26.3	89.6	+63.3	HIGH
Machine Learning Engineer	23.8	61.2	+37.4	HIGH
ML Engineer	22.6	53.3	+30.6	HIGH
Data Scientist	15.6	61.7	+46.1	HIGH
MLOps Engineer	12.9	56.3	+43.4	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

System Design

3 of 6 roles

systems

Business Communication

1 of 6 roles

business

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

1 **HIGH** **Add System Design coverage** — required vocabulary in 3 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.

2 **LOW** **Optional: Business Communication** — would broaden appeal to one additional role. Candidate terms: business.

GAP KEYWORDS BY ROLE

AI Engineer

27.8 → 66.1 (+38.3 pts)

perception data learning build software technical code model

Applied Scientist

26.3 → 89.6 (+63.3 pts)

discovery scientific research interdisciplinary technical data learning systems

Machine Learning Engineer

23.8 → 61.2 (+37.4 pts)

data predictive learning analytics tools solutions exploratory business

ML Engineer

22.6 → 53.3 (+30.6 pts)

discovery data learning machine systems model build new

Data Scientist

15.6 → 61.7 (+46.1 pts)

toolkit definition data analytics scripting mitigation social build

MLOps Engineer

12.9 → 56.3 (+43.4 pts)

data model scientists systems learning machine tools development

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.